

Published as Geophysics, vol 85, issue 5, O71-O82 (2020)

Quantifying and correcting residual azimuthal anisotropic moveout in image gathers using dynamic time warping

Luke Decker  and Qunshan Zhang 
 *The University of Texas at Austin*
 Repsol USA

Abstract:

We propose and demonstrate a novel application of dynamic time warping (DTW) for correcting residual moveout in image gathers, enhancing seismic images, and determining azimuthal anisotropic orientation and relative intensity when moveout is caused by wave propagation through a media possessing elliptical horizontally transverse isotropy (HTI). The method functions by first using DTW to determine the sequences of integer shifts that most closely match seismic traces within an image gather to the gather's stack, and then applying those shifts to flatten the gather. Flattening shifts are fitted to an ellipse to provide an approximation for the orientation and relative strength of elliptical HTI anisotropy. We demonstrate the method on synthetic and 3D field data examples to show how it is able to (1) correct for residual azimuthal anisotropic moveout, (2) accurately recover high frequency information and improve feature resolution in seismic images, and (3) determine the anisotropic orientation while providing a measure of relative strength of elliptic anisotropy. We find that while the method is not intended to replace anisotropic processing techniques for moveout correction, it has the ability to inexpensively approximate the effects of such operations while providing a representation of the elliptic HTI anisotropy present within a volume.

- [Introduction](#)
- [Theory](#)
- [Synthetic Gather Example](#)
- [Field Data Example](#)
- [Discussion](#)
- [Conclusions](#)
- [Acknowledgments](#)
- [Bibliography](#)
- [About this document ...](#)

论文结构